

KATAKAM SAIKUMAR

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Professional Summary

Software Development Engineer with an MS in Information Systems from Northeastern University (GPA: 3.6/4.0, May 2026) and 1+ years of experience building **scalable, fault-tolerant distributed systems** using **Java, Python, and C++**. Proven track record in **cloud-native architectures on AWS, microservices, CI/CD pipelines**, and **Agile** delivery. Strong foundation in **Data Structures, Algorithms, OOP, and System Design**; experienced with **GenAI/AI-powered tools** to boost development productivity. Passionate about **customer-obsessed, production-grade software** with end-to-end ownership from design through deployment and operations.

Education

Northeastern University

Boston, MA

Master of Science in Information Systems; GPA: 3.6/4.0

Aug 2024 – May 2026

- **Coursework:** Data Structures & Algorithms, Cloud Computing, Distributed Systems, Machine Learning

SRM Institute of Science and Technology

Chennai, India

Bachelor of Engineering in Electronics and Communications; GPA: 3.6/4.0

Aug 2020 – May 2024

Professional Experience

Software Development Engineer

Nov 2023 – Jun 2024

Eastencher Software Private Limited

Bengaluru, India

- Designed and deployed **10+ RESTful APIs** using **Java Spring Boot** following **OOP** and **SOLID** principles across **microservices**, reducing inter-service latency by **25%**; participated in **code reviews** and maintained technical documentation for cross-team alignment
- Optimized **MySQL** schemas with indexing and query tuning, improving read performance by **35%**; implemented **unit and integration tests** using **JUnit** achieving **80%+ code coverage** for production-grade reliability
- Owned end-to-end feature delivery in **Agile/Scrum** sprints using **Git, Docker**, and **CI/CD pipelines** via **GitHub Actions**, cutting release cycle time by **30%**; monitored and troubleshooted production issues to ensure operational excellence

Software Engineer – Data & ML

Aug 2023 – Nov 2023

Quicknify Software Technologies

Bengaluru, India

- Built scalable **Python** data pipelines processing **100K+ records/day**, optimizing **SQL queries** via **hash map** and **tree-based indexing** strategies, reducing end-to-end pipeline runtime by **40%**
- Designed modular **ETL architecture** using **Python (OOP)** and **PostgreSQL** with automated data validation, maintaining **99.5% data accuracy** across distributed datasets; leveraged **AI-assisted development tools** to accelerate delivery

Technical Projects

NextGen Agriculture Platform | Java, Spring Boot, React, REST APIs, PostgreSQL, Microservices, Docker

2026

- Architected a **full-stack distributed platform** with **6 microservices** across financial, supply, and logistics domains; designed **RESTful APIs** with **role-based access control (RBAC)** supporting **9 user roles** and multi-org **PostgreSQL** schemas with optimized joins
- Containerized with **Docker**; automated deployments via **CI/CD (GitHub Actions)**; integrated real-time weather APIs and participated in structured **code reviews** to maintain clean, maintainable code

Fault-Tolerant Streaming Data Pipeline | Apache Kafka, Spark Streaming, Python, AWS S3, Redshift

2026

- Built **cloud-native streaming platform** on **AWS** processing **100K+ events/hour** with **Apache Kafka** and **Spark Streaming**; enforced **exactly-once semantics** with sub-**30-second** end-to-end latency and built-in fault tolerance
- Engineered partitioned **data lake on AWS S3 (Parquet)**; designed **star schema** in **Redshift** with automated incremental loading; **unit tests** reduced data quality issues by **35%**

Live Stock Market Dashboard | Python, FastAPI, React, WebSocket, AWS EC2, Docker, CI/CD

2025

- Developed **full-stack real-time dashboard** with **Python FastAPI** and async **WebSocket** architecture serving live market data to a **React** frontend with low-latency refresh; deployed on **AWS EC2** with **Docker** and **GitHub Actions CI/CD**
- Achieved **85%+ code coverage** with **pytest**; monitored application health post-deployment, demonstrating **operational excellence** and ownership across the full software development lifecycle (SDLC)

Technical Skills

Languages: Java, Python, JavaScript/TypeScript, C, C++, SQL

Frameworks & Libraries: Spring Boot, FastAPI, Flask, React, Node.js, Apache Kafka, Spark Streaming, Pandas, NumPy, scikit-learn

Cloud & DevOps: AWS (EC2, S3, Lambda, Redshift), Docker, Git, GitHub Actions (CI/CD), GenAI/AI Dev Tools

Databases: PostgreSQL, MySQL, MongoDB, Redis

Concepts: Distributed Systems, Microservices, System Design, OOP, DSA, REST APIs, Agile/Scrum, CI/CD, Unit Testing, Operational Excellence